



Figure 1: Reviewing hardware and software requirements

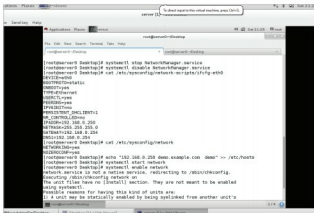


Figure 2: Static network services and network manager disabled

The IDm server is supported on both i386 and x86_64 platforms of Red Hat Enterprise Linux 6. Properly sizing the RAM is imperative given the caching philosophy of IDm, i.e., at least 2GB RAM for 10,000 users and at least 16GB of RAM for 100,000 users and 50,000 groups. The installation script makes certain assumptions about the systems, i.e., DNS and */etc/hosts* properly resolve the fully-qualified domain name, there is no pre-existing 389 Directory Server or Kerberos configuration, and the NSCD (Name Service Cache Daemon) service is disabled. System ports for IDm services include HTTP/HTTPS (80/tcp and 443/tcp), LDAP/ LDAPS (389/tcp and 636/tcp), Kerberos (88 and 464, both tcp and udp), DNS (53, both tcp and udp), NTP (123/udp), and Certificate System (7389/tcp). Network configuration should disable NetworkManager (and NetworkManagerDispatcher if installed), and use static network services.

2. Next, install the IdM server packages. Installing `ipa-server` will typically install a large number of dependencies. If the IdM server also manages a DNS server, then you must specifically install two additional packages — `bind` and `bind-dyndb-ldap`.

```
# yum -y install ipa-server.
```

- Now, create an IdM server instance interactively. The

ipa-server-install script will configure all of the required services for the IDM domain — network time domain (ntpd), 389 directory server (dirsrv), Kerberos key distribution center (KDC), Apache (httpd), certificate authority, and an updated SELinux targeted policy. Optionally, it can integrate a DNS service. You can monitor the installation from the `/var/log/ipaserver-install.log` log file. Each component service (Apache, 389 Directory Server, Kerberos, and Certificate System) has its own log files, generally found under `/var/log/` (`httpd/`, `dirsrv/`, `k/`, `log`, and `pki-ca/`).

```
#ipa-server-install.
```

The same result could be achieved by passing the following required arguments to the `ipa-server-install` command.

```
# ipa-server-install -hostname=demo.example.com -n example.com  
-r EXAMPLE.COM -p linux123 -a redhat123 -U
```

⚠ Warning: The hostname must be a valid DNS name—only numbers, lower-case letters of the alphabet and hyphens. It cannot use underscores or capital letters.

Unlike Active Directory, where AD DNS must equal the AD domain, with IdM, a single deployment (Kerberos RELAM) can manage several domains without duplication of the servers. This is why we specify the `-n` DNS domain and `-r` Kerberos RELAM separately.

The `-p` option sets the underlying directory manager password, which is used for troubleshooting functions. Think of it as the root of the IdM LDAP instance, and it should be protected as such. The `-a` option sets the admin user password, which represents the first administrative account and is used for general IPA administrative tasks. Other options include `-setup-dns` to integrate the DNS service and `-idmax=number` and `-idstart=number`, which can be used to bind the IDs assigned by the IdM server. If the server is being installed on a virtual machine, it is normally recommended not to run an NTP server. To disable NTP for IdM, use the `-no-ntp` option.

- Restart the ssh service to obtain Kerberos' credentials.

```
# systemctl start sshd.
```
- Verify the IdM instance by verifying Kerberos' authentication and ipa command line access. Verify Kerberos authentication.

Adding user and group entries to IdM

1. Before we use the command line tool for the Web based interface, first log in to the IdM domain to obtain a Kerberos ticket with administrative authority.
 2. The command line utility, *ipa*, offers a number of global options.
- The *ipa* command supports a number of sub-commands,